Midterm 2 Partial Credit

On problem 19, if I knew that A^2 is different from A@A in numpy, I think that I would have selected the correct answer. In my work, I did make an additional error (I didn't include C2@A2^3), but I'm confident that I would have figured that out if my third row had matched with the last row all zeros. I also know however, that there is a post that I did not look at before the test on Piazza explaining this exact issue, so I understand if partial credit is not given.

Original work:

19. (5 points) Given the following state space model, find the observability matrix for a disturbance observer:

$$\dot{x} = \begin{pmatrix} -1 & 2 \\ 1 & 0 \end{pmatrix} x + \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} u \tag{6}$$

$$y = \begin{pmatrix} 1 & -1 \end{pmatrix} x. \tag{7}$$

(a) $\begin{pmatrix} 1 & -3 & 5 & -11 \\ -1 & 1 & -3 & 5 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}$

(b) $\begin{pmatrix} 1 & -1 & 0 & 0 \\ -2 & 2 & 1 & -1 \\ 4 & -4 & -2 & 2 \\ -8 & 8 & 4 & -4 \end{pmatrix}$

(c) $\begin{pmatrix} 1 & -3 \\ -1 & 1 \end{pmatrix}$

(d) $\begin{pmatrix} 1 & -1 \\ -2 & 2 \end{pmatrix}$

(e) None of the above.

A_2 must be square! (2 inputs)

$$A_2 = \begin{bmatrix} \begin{bmatrix} -1 & 2 \\ 1 & 0 \end{bmatrix} & \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \\ \begin{matrix} 0 & 0 \\ 0 & 0 \end{matrix} & \begin{matrix} 0 & 0 \\ 0 & 0 \end{matrix} \end{bmatrix}$$

$$C_2 = [1 \quad -1 \quad 0 \quad 0]$$

$$\mathcal{O}_{A_2, C_2} = \begin{bmatrix} C_2 \\ C_2 A \\ C_2 A^2 \end{bmatrix} = \begin{bmatrix} 1 & -1 & 0 & 0 \\ -2 & 2 & 1 & -1 \\ 0 & 4 & 1 & -1 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

NOT observable

```python
def Q19():
    print("Q 19 ================================")
    A = np.array([[-1, 2],
                  [1, 0]])
    B = np.array([[1, 0],
                  [0, 1]])
    C = np.array([[1, -1]])

    inputs = B.shape[1]
    A2 = np.block([[A, B],
                   [np.zeros((inputs, A.shape[1])), np.zeros((inputs, inputs))]])
    C2 = np.block([[C, np.zeros((C.shape[0], inputs))]])
    O_A2C2 = np.block([[C2], [C2@A2], [C2@A2**2]])
    print(f"{O_A2C2 = }")
```

Corrected work:

```python
def Q19():
    print("Q 19 ==================================")
    A = np.array([[-1, 2],
                  [1, 0]])
    B = np.array([[1, 0],
                  [0, 1]])
    C = np.array([[1, -1]])

    inputs = B.shape[1]
    A2 = np.block([[A, B],
                   [np.zeros((inputs, A.shape[1])), np.zeros((inputs, inputs))]])
    C2 = np.block([[C, np.zeros((C.shape[0], inputs))]])
    O_A2C2 = np.block([[C2], [C2@A2], [C2@A2@A2], [C2@A2@A2@A2]])

    print(f"{A2 = }")
    print(f"{C2 = }")
    print(f"{O_A2C2 = }")
```

```
$ python midterm2.py
A2 = array([[-1.,  2.,  1.,  0.],
       [ 1.,  0.,  0.,  1.],
       [ 0.,  0.,  0.,  0.],
       [ 0.,  0.,  0.,  0.]])
C2 = array([[ 1., -1.,  0.,  0.]])
O_A2C2 = array([[ 1., -1.,  0.,  0.],
       [-2.,  2.,  1., -1.],
       [ 4., -4., -2.,  2.],
       [-8.,  8.,  4., -4.]])
```

Which matches B as the correct answer.